

CS106S Course Calendar

This is a coarse sketch of topics we'll talk about over the course of the quarter, though I can't promise I'll stay true and loyal to everything here ☺. I have trouble keeping a consistent bedtime or a remotely organized wardrobe, never mind an ever-evolving syllabus for many brilliant students—designed so that we include topics that are relevant, accessible, and thought-provoking to each of you, and broadly to the planet.



Nevertheless, I think it's reasonable to give you an approximate outline (based on what's been really taught in the past), as you decide if you'd like to take CS106S or not. No hard feelings if you drop. I, in fact, dropped this class as a frosh years ago oops, and never ended up taking it. Yikes. Life is a movie.

Lastly, if any topics seem particularly exciting or terribly boring, or if there's anything you'd like to add, reach out after class, and I'd love to chat. Given how freewheeling this course is, we can probably do it! Some vague topics I've been floating for years—education, politics, law, urban design, accessible tech.

Week 1: Intro for CS for Social Good, JavaScript, and Cryptography

Sept 24th

Beyond the intro remarks and introducing ourselves, we'll have a relaxed primer on the JavaScript language—which I promise is quite beautiful, artistic, and elegant, despite the many Internet memes on its most bizarre features, e.g., `[1,2] + [3,4] = "1,23,4"`.

The short story is that JavaScript, like Python, C++, Java, supports numbers, strings, arrays, and maps. JavaScript, however, uses one keyword, `let`, to declare variables of any type. For those who've programmed with another language before, this will likely feel more like a transition to new syntax than having to acquire CS concepts from scratch. As you power through the starter exercises, I'll circle around the room for folks with any questions or rightful complaints. Our end goal for the day will be implementing a Caesar cipher—and if we have time, a garden variety of optional extension exercises. Lastly, I'll bring mochi ☺.

Week 2: Sentiment Analysis and Monitoring Refugee Tweets

Oct 1st

We'll discuss algorithms for sentiment analysis—the tricky process of identifying whether a piece of text espouses a positive, negative, or likely neutral tone, from movie reviews to political opinion & polarization. Accordingly, we'll write and evaluate a program that flags harmful tweets regarding refugees, by quantifying the sentiment of individual words. We'll also glimpse modern approaches to sentiment analysis such as word embeddings/vectors.

Week 3: CS for Climate Change and Google Earth Engine

Oct 8th

We'll learn about a wondrous web program you've likely never heard of before: the Google Earth Engine, which allows to render the entire planet in blazing lightspeed at your fingertips. Using its JavaScript API, we'll analyze interactive satellite imagery, with a focus on effective color visualization and the sobering reality of worldwide deforestation and habitat destruction. Next, you'll create your own image visualization of any kind, drawing from any resources—and if you'd like, share it with us in a class-wide art gallery! (of sorts)

- Week 4:**
Oct 15th **Disease Detection, K-Nearest Neighbors Algorithm, K-Means Clustering**
We'll introduce basic machine learning concepts (e.g., clustering, classification) in a way such that prior experience is not only unnecessary, but almost spoils the novelty. From there, we'll learn about a duet of classic, surprisingly potent machine learning algorithms: K-Nearest Neighbors and K-Means clustering, which can each be implemented in no more than 20 lines of raw JavaScript. After coding them up, we'll apply the algorithms to classify cell samples as either benign or malignant, and discuss the sensitive implications.
- Week 5:**
Oct 22nd **Cybersecurity and Ethical Web Hacking**
We'll (legally) hack a real website, and learn about all kinds of security vulnerabilities and how to exploit them (ethically, of course ☺, e.g., to allow developers to safeguard their systems better). Attacks that we'll discuss include IDOR, e.g., URL manipulation to access private data, XSS to run malicious JavaScript code in someone's browser (i.e. like a Trojan horse) and, à la Santa Claus, stealing (web) cookies to log into other peoples' accounts.
- Week 6:**
Oct 29th **Rethinking the Electoral College, Alternative Voting Methods**
As the week before the pivotal 2026 U.S. midterm elections, you may very well have the dire state of modern-day politics on your mind. Using heuristic methods and/or recursion, we'll estimate the lowest percentage of votes one needs to become U.S. President via the Electoral College (it's comically low). We'll discuss and maybe even implement alternate voting methods, such as ranked-choice voting, or just electing who gets the most votes.
- Week 7:**
Nov 5th **Internet Trust and Safety**
Many people I've met in Trust and Safety teams have adopted a truly morbid, cartoonishly tragic sense of comedy to cope with their work, which is investigating the worst parts of the Internet—and how people can abuse it to do serious harm. We'll deliberate real-world case studies ranging from YouTube to Google Search to dating apps. Our code exercise focuses on modifying a JavaScript-powered mini search engine to steer users away from extremist and harmful content, using the "Redirect Method" pioneered at Google Jigsaw.
- Week 8:**
Nov 12th **Mental Health, ELIZA Chatbot, Campus Well-Being**
Chatbots are everywhere now, thanks to ChatGPT. But instead of engaging with GPT, Claude, Gemini, Spicy Pisces, whatever, we'll do something a little unexpected, and take a time machine all the way back to the first ever one, ELIZA. This chatbot (1966) uses clever pattern matching and string substitution to emulate a mental health therapist with an attitude. You'll receive a carbon copy of its recreated source code—in JavaScript ☺—to test out conversations on your own, as well to brainstorm and make direct, code-level improvements. We'll end by discussing mental health resources on Stanford's campus 🤖.
- Week 9:**
Nov 19th **What's Next?, Stanford CS Stories, and End-Quarter Boba Party 🍷**
Come with questions, an open mind, a strong-willed heart, and anything you'd like to share! I'll even talk about my personal, wacky, outrageous CS story that I won't even pretend is anything inspiring lol. To quote Iron Man, "don't do anything I would do, and definitely don't do anything I wouldn't do." ☺ And of course, I'll send out an order form ahead of time for free boba, paid for by the CS department, who probably hates my guts by now. I'll also invite some fellow Stanford uncs I know with colorful adventures to tell.
- Week 10:** **No class, best of luck on finals! 🍀**